



Capstone Project 2

CMU-SE 451

User Interface Design Document

Version 1.0

Date: 15 - Feb - 2024

Sign Language Detection System

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Date

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Revision history

Version	Date	Description of Changes	Author
1.0	15/02/2024	Initial document	Team

1. Project Overview

PROJECT INFORMATION

Project acronym	SLDS		
Project Title	Sign Language Detection System		
Start Date	15 - Feb - 2024	End Date	20 - May - 2024
Lead Institution	International School, Duy Tan University		
Project Mentor	Name: Thuan, Nguyen Tan Ph.D. Email: thuannt.it.dtu@gmail.com Tel: 0905.626.276		
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2. Introduction

2.1. Purpose

- This document serves the purpose of specifying application's UI design

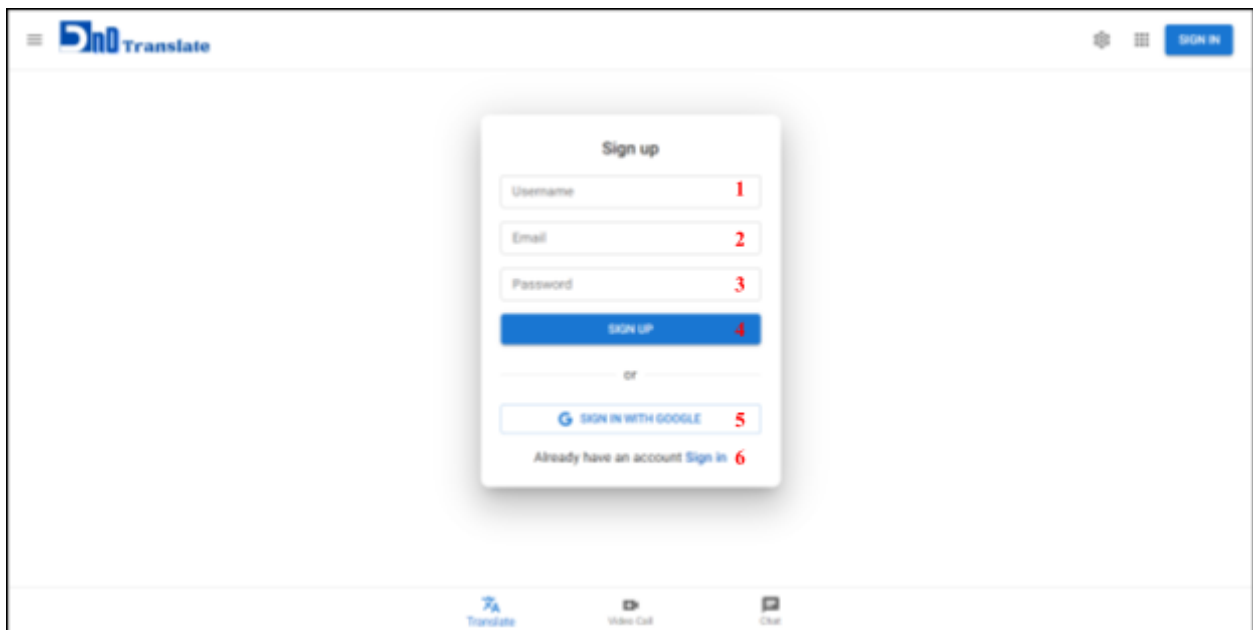
2.2. Scope

- Create complete interfaces, and describe them clearly.
- Overview of the stages and modules of the system

3. Describe User Interface Application in Detail

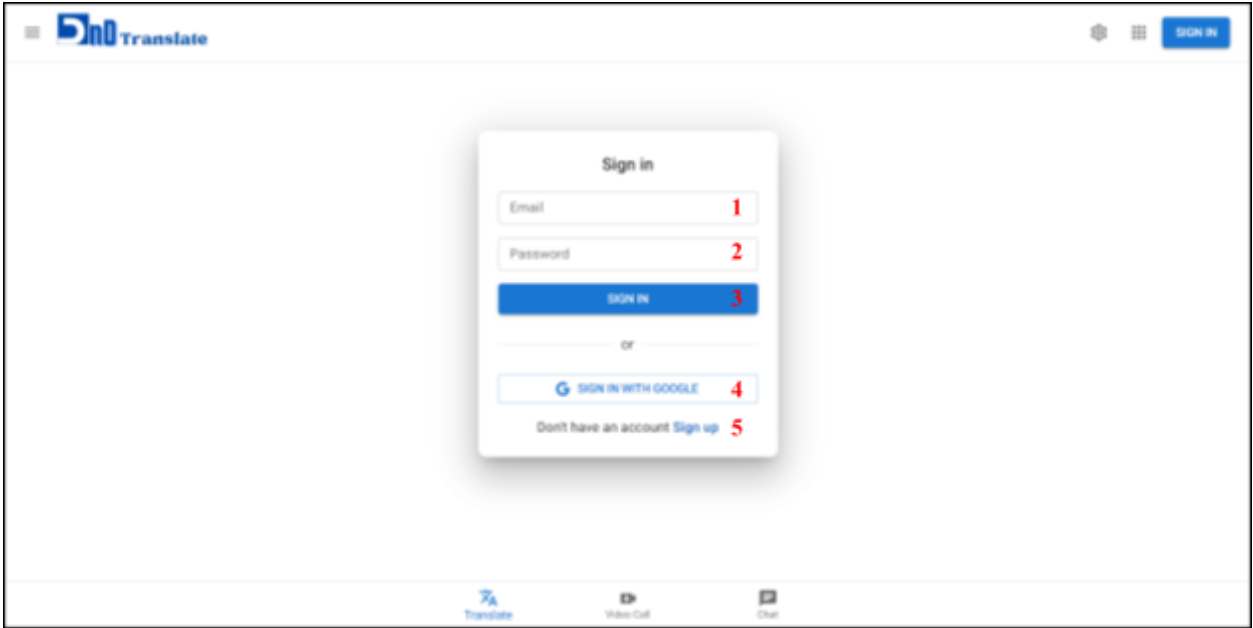
3.1. User Authentication and Account Management

3.1.1. Sign up



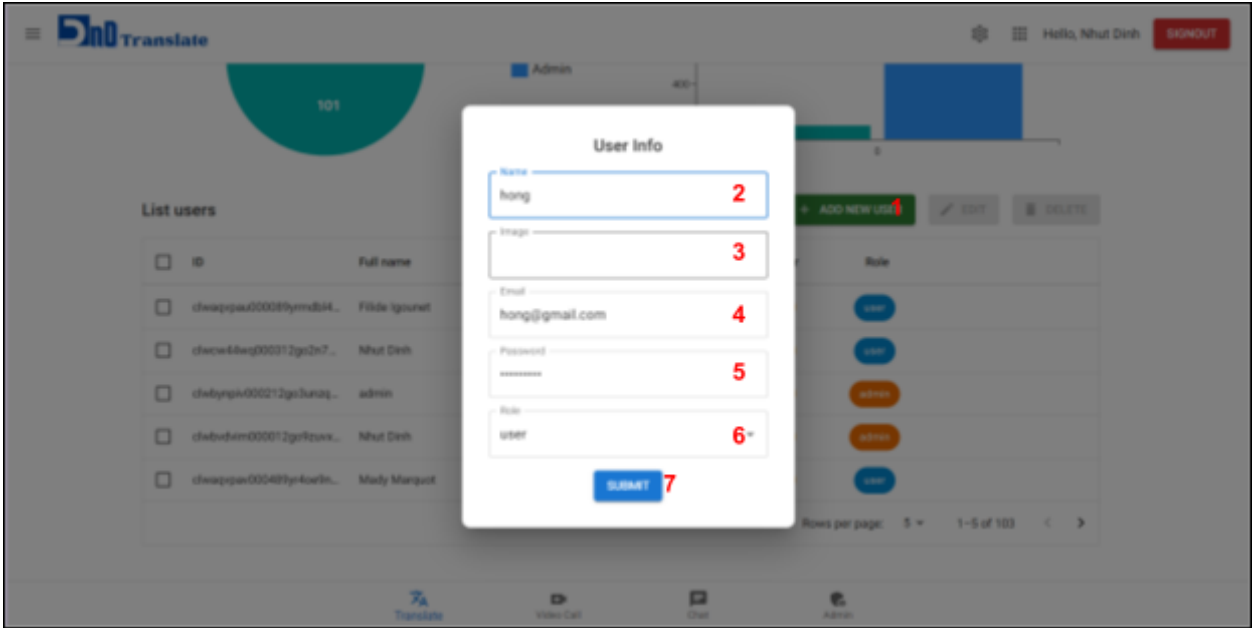
ID	Condition	Control Type	Target	Notes
1	Click	TextField	Fill username	
2	Click	TextField	Fill email	
3	Click	TextField	Fill password	
4	Click	Button	Click to submit sign up with credentials	
5	Click	Button	Sign in with Google account	
6	Click	Link	Redirect to Sign in page	

3.1.2. Sign In



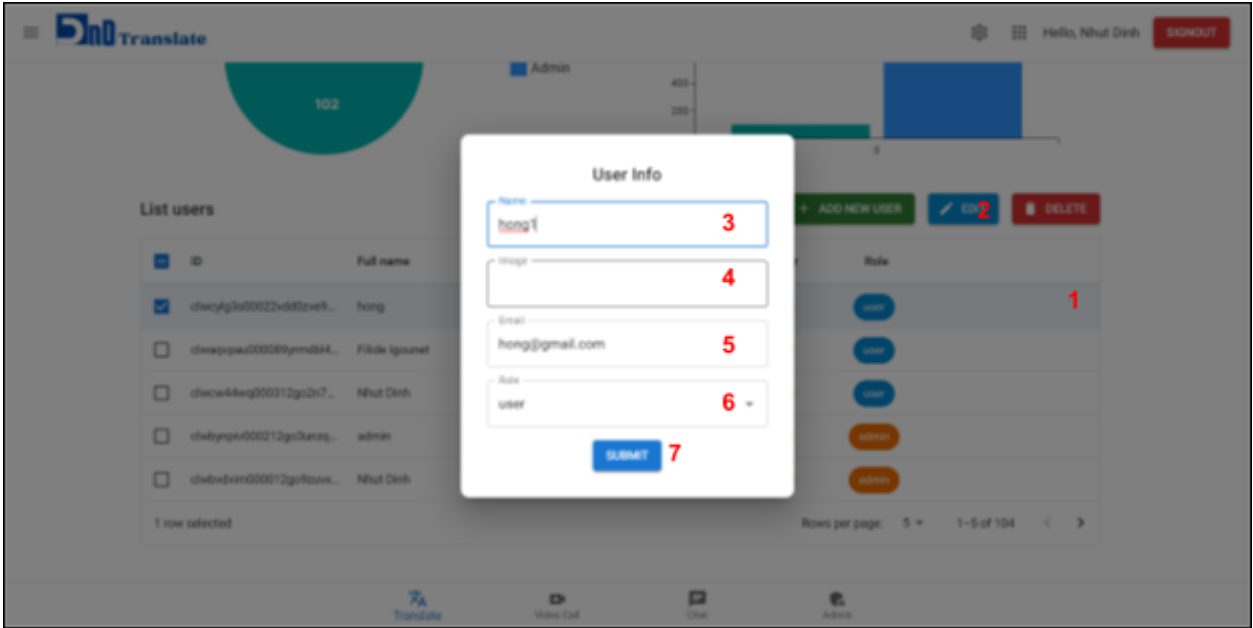
ID	Condition	Control Type	Target	Notes
1	Click	TextField	Fill email	
2	Click	TextField	Fill password	
3	Click	Button	Click to submit sign in with credentials	
4	Click	Button	Sign in with Google account	
5	Click	Link	Redirect to Sign up page	

3.1.3. Create new user



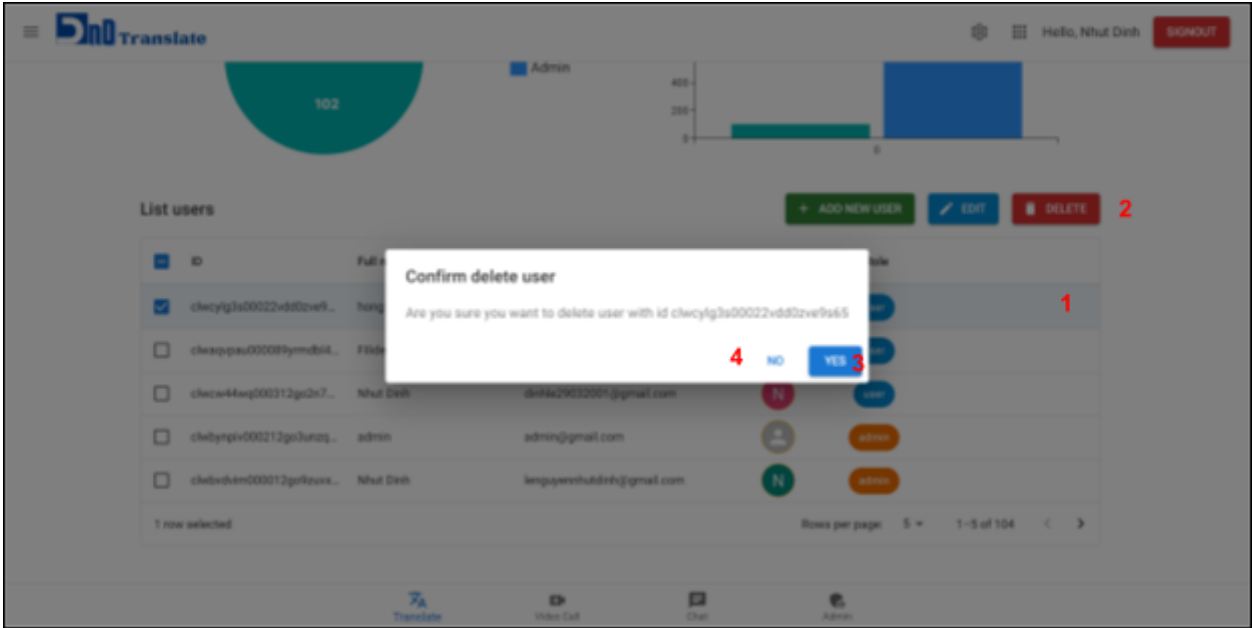
ID	Condition	Control Type	Target	Notes
1	Click	Button	Click button to open create user form	
2	Click	TextField	Fill name	
3	Click	TextField	Fill image	
4	Click	TextField	Fill email	
5	Click	TextField	Fill password	
6	Click	DropDownList	Choose role	
7	Click	Button	Click to submit create user form	

3.1.4. Edit user



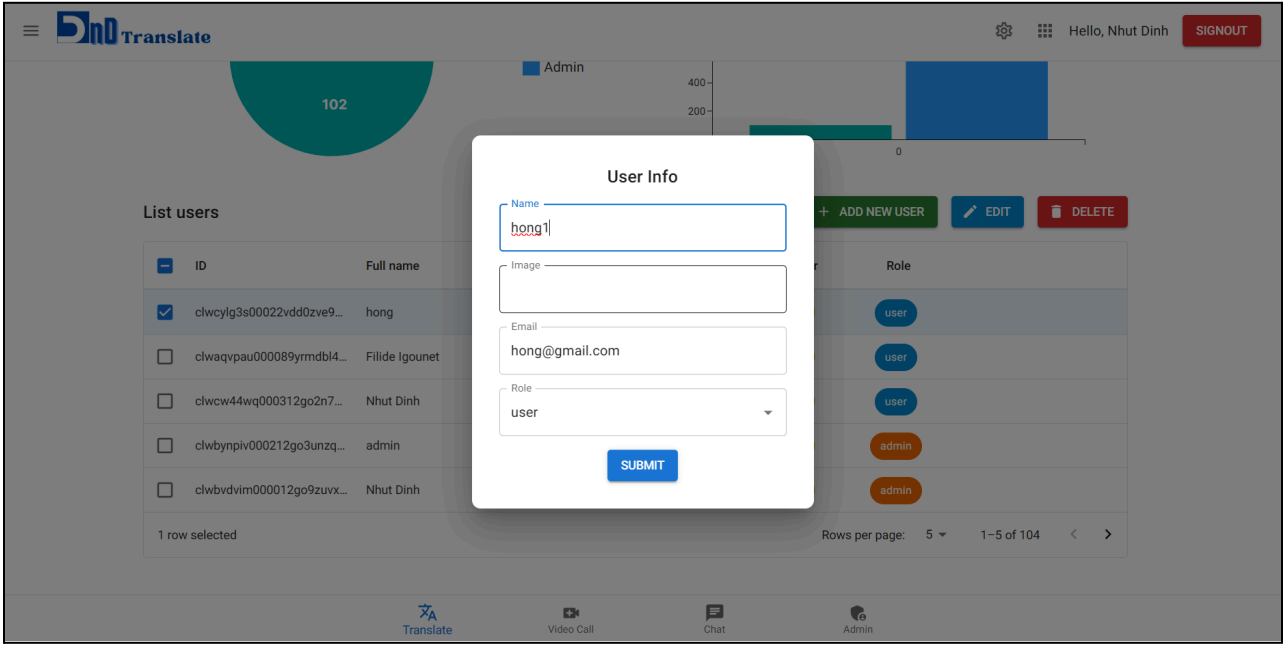
ID	Condition	Control Type	Target	Notes
1	Click	RowData	Choose user row	
2	Click	Button	Click button to open edit user form	
3	Click	TextField	Fill name	
4	Click	TextField	Fill image	
5	Click	TextField	Fill email	
6	Click	DropDownList	Choose role	
7	Click	Button	Click to submit edit user form	

3.1.5. Remove user



ID	Condition	Control Type	Target	Notes
1	Click	RowData	Choose user row	
2	Click	Button	Click button to open dialog confirm	
3	Click	Button	Click “Yes” to delete user	
4	Click	Button	Click “No” to cancel	

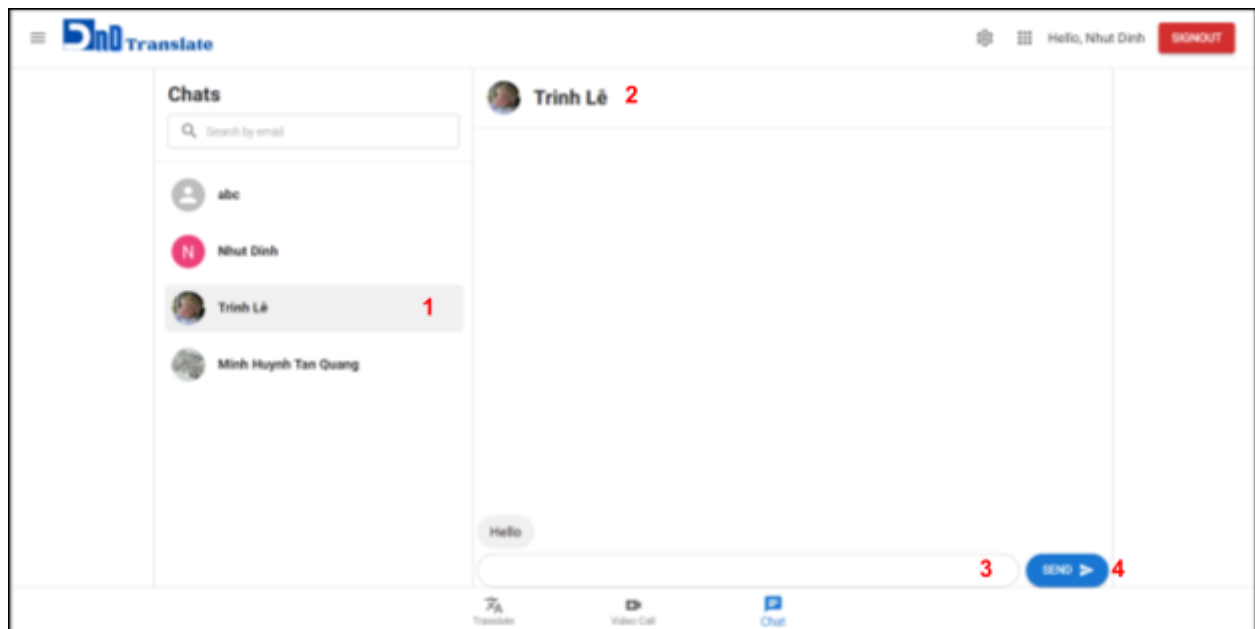
3.1.6. Detailed user profile



ID	Condition	Control Type	Target	Notes
1	Click	RowData	Choose user row	
2	Click	Button	Click button to open detail user form	
3	Click	TextField	Show name	
4	Click	TextField	Show image	
5	Click	TextField	Show email	
6	Click	DropDownList	Show role	

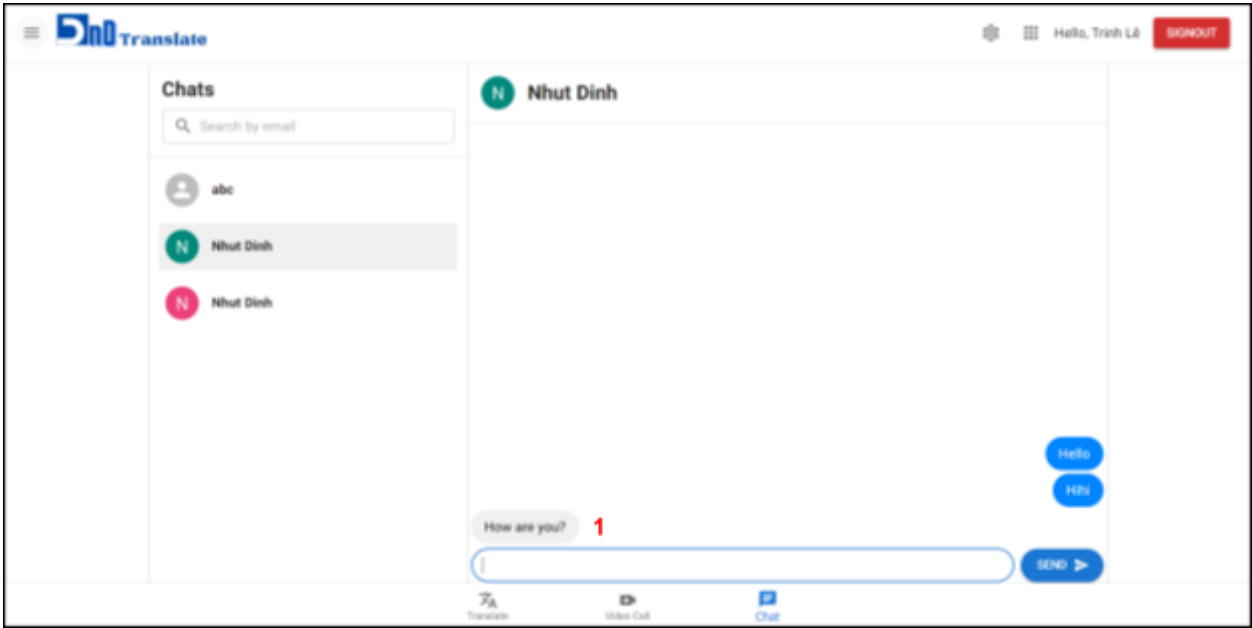
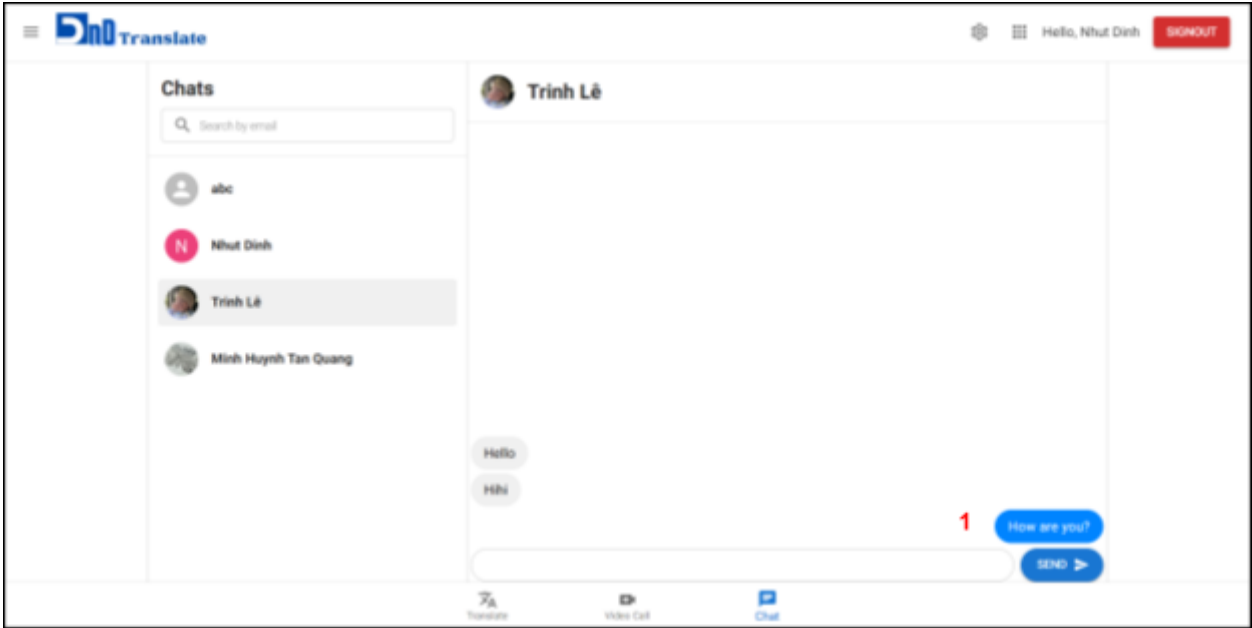
3.2. Communication and Messaging

3.2.1. Add messages



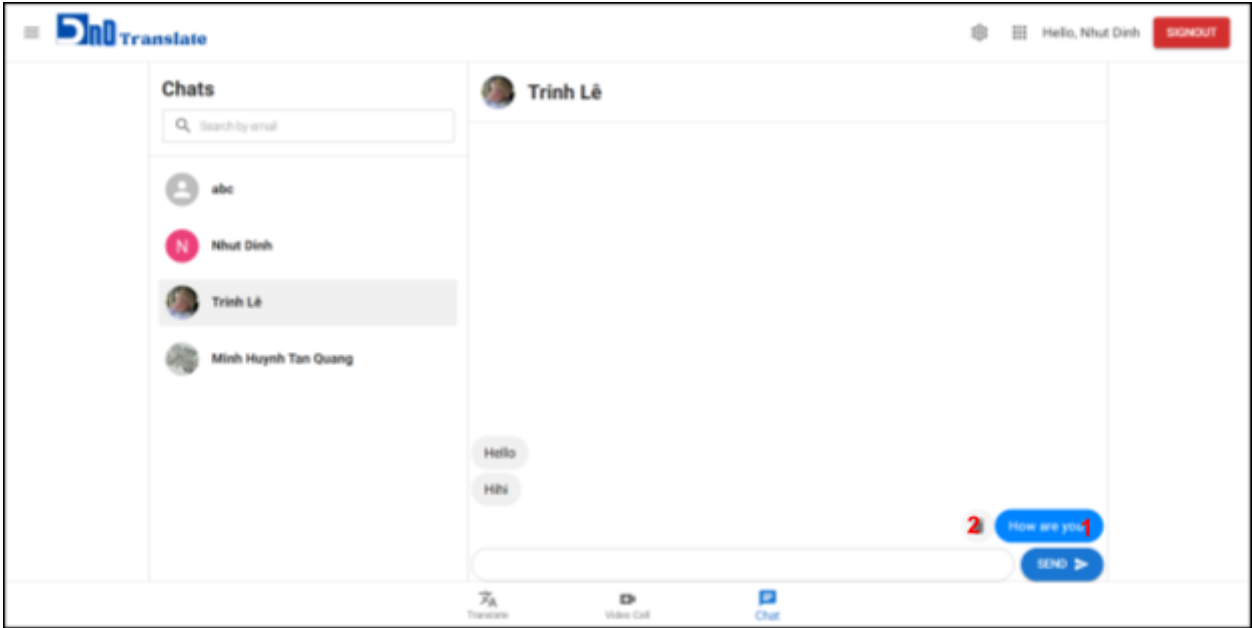
ID	Condition	Control Type	Target	Notes
1	Click	ListBox	Open conversation of user clicked	
2	Nothing	Label	Show avatar and name of the friend	
3	Click	TextField	Fill message content want send	
4	Click	Button	Send message content to the friend	

3.2.2. Realtime messages



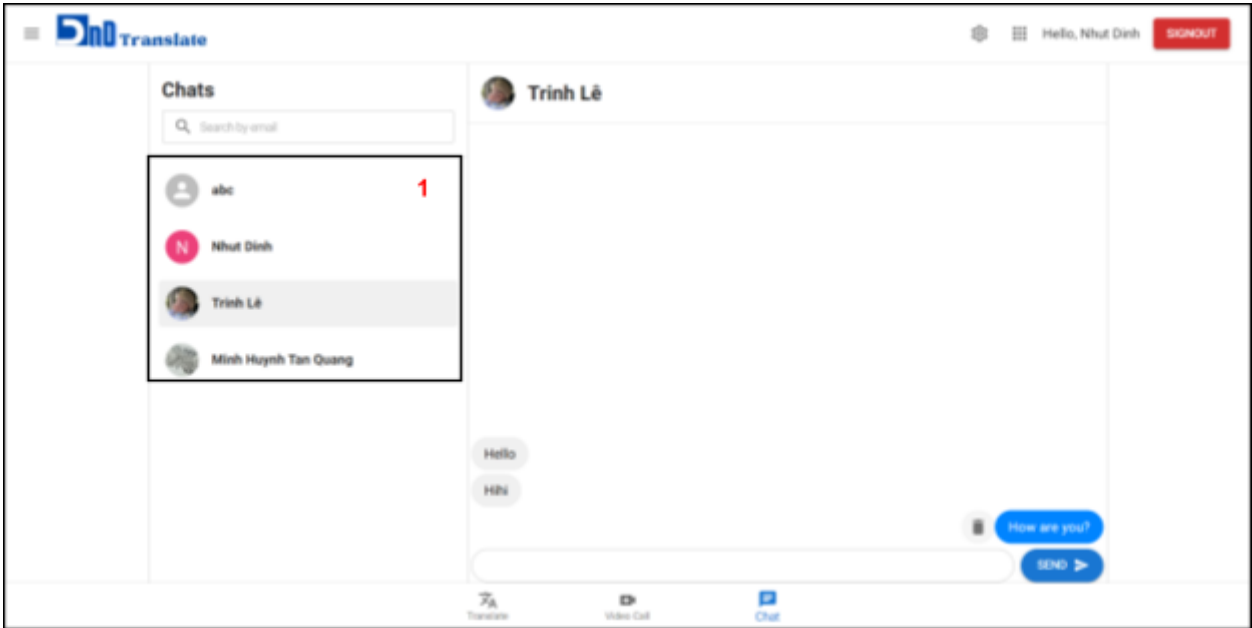
ID	Condition	Control Type	Target	Notes
1	Nothing	Label	Sync messages content instantly	

3.2.3. Delete message



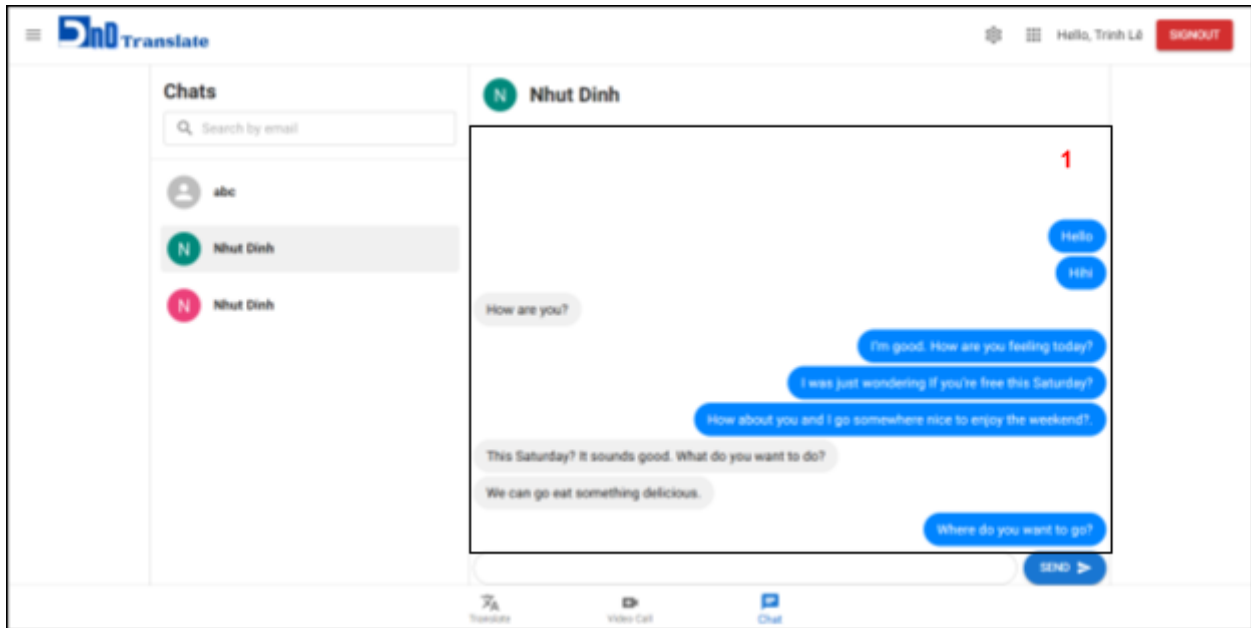
ID	Condition	Control Type	Target	Notes
1	Hover	Label	Show delete icon	
2	Click	Button	Click button to delete message	

3.2.4. List all conversations



ID	Condition	Control Type	Target	Notes
1	Nothing	ListBox	Show list conversations	

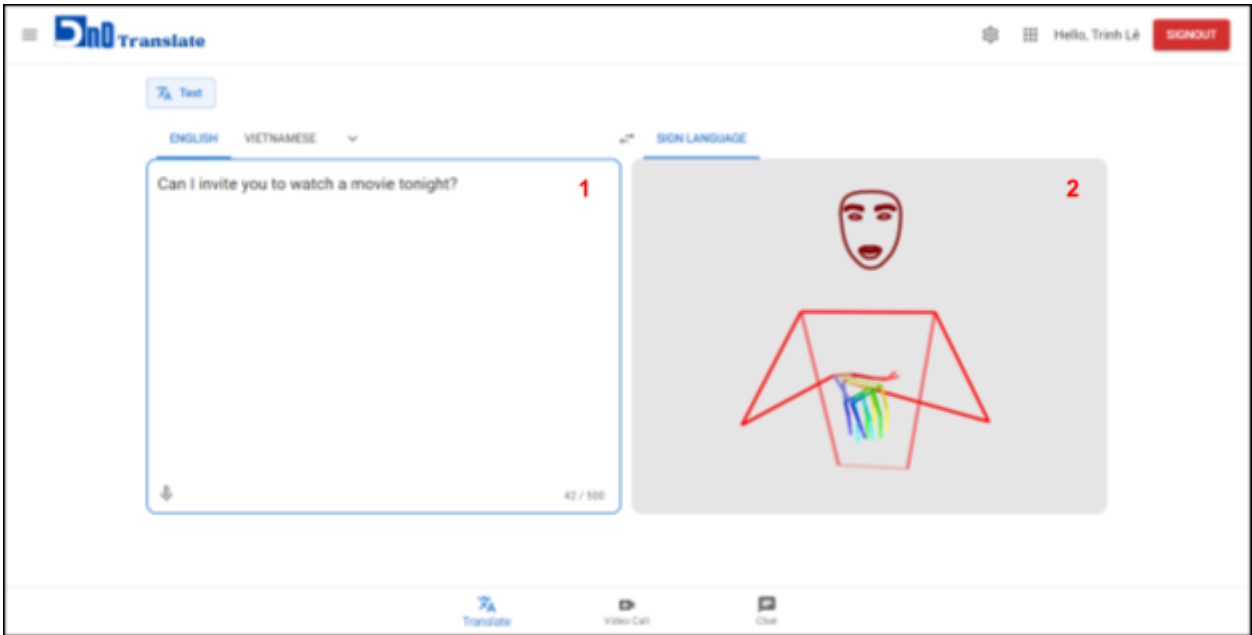
3.2.5. List all messages within a conversation



ID	Condition	Control Type	Target	Notes
1	Nothing	Label	Show list messages	

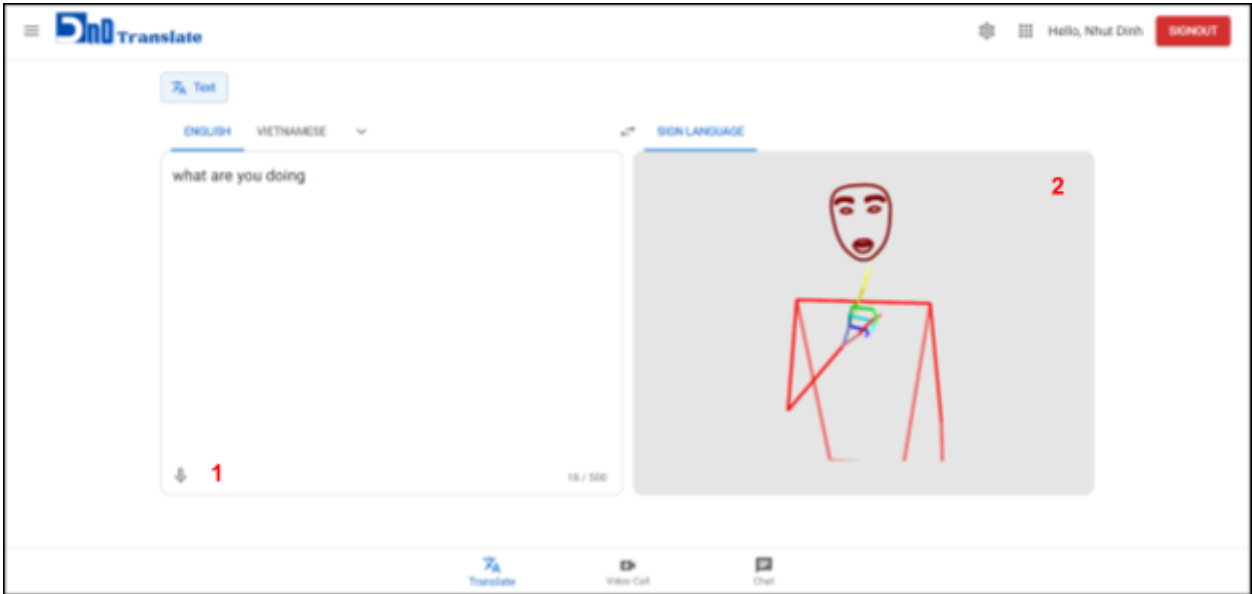
3.3. Sign Language Recognition

3.3.1. Convert text to sign language



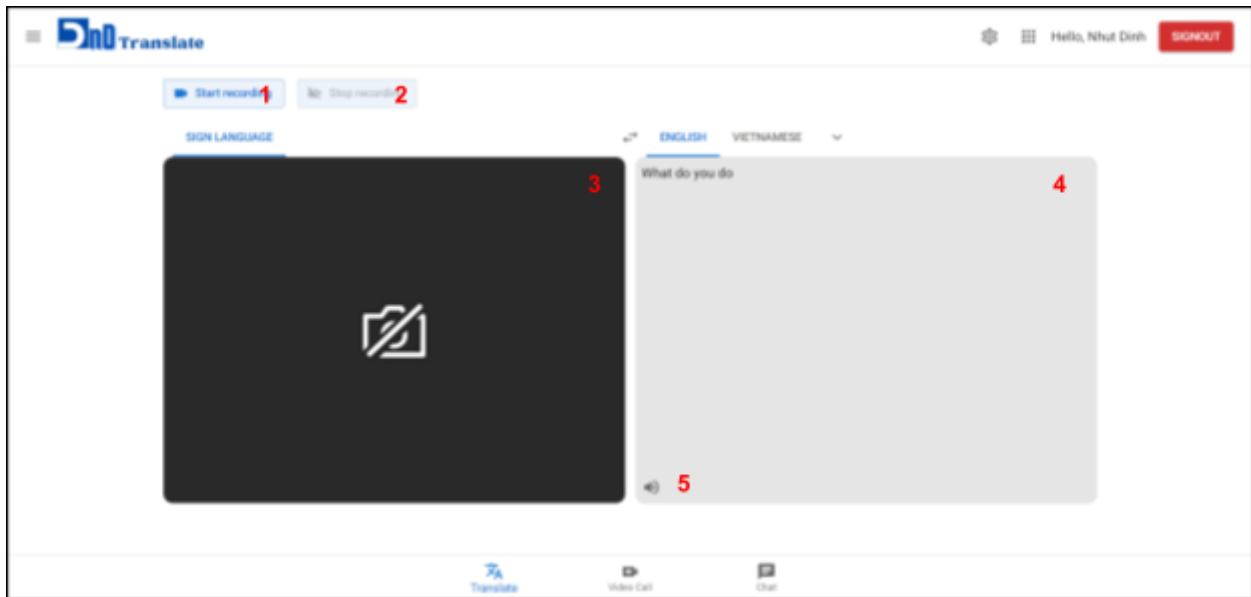
ID	Condition	Control Type	Target	Notes
1	Click	TextField	Fill text	
2	Nothing	SignLanguageModel	Show sign language	

3.3.2. Convert voice to sign language



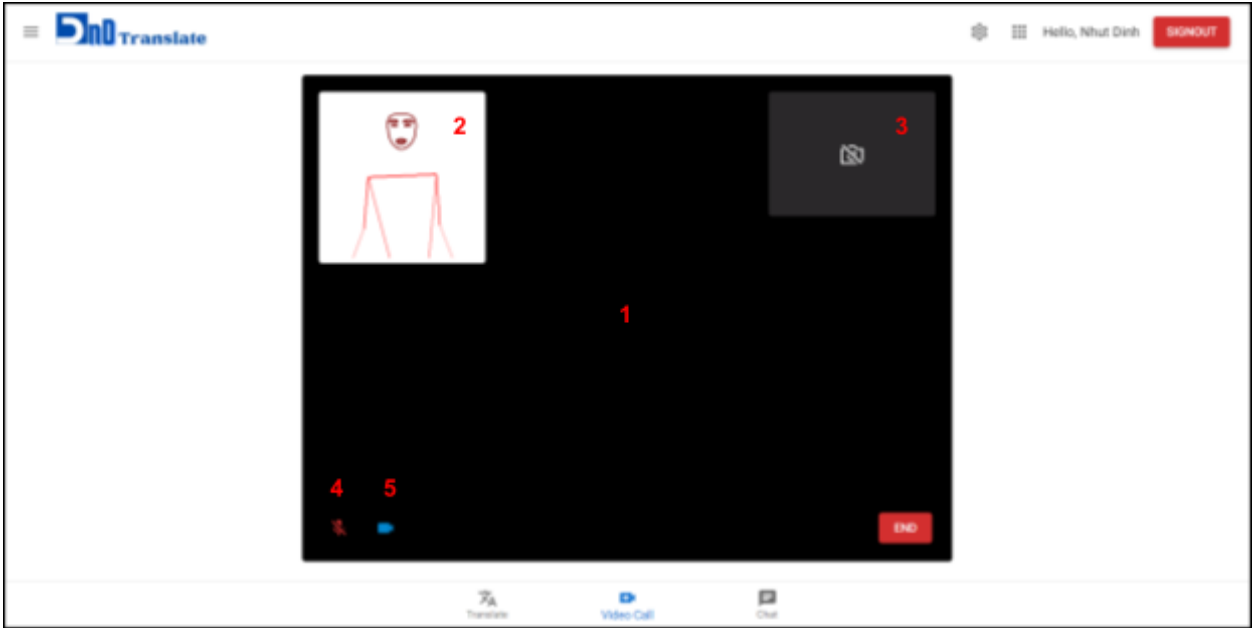
ID	Condition	Control Type	Target	Notes
1	Click	Voice	Speak	
2	Nothing	SignLanguageModel	Show sign language	

3.3.3. Convert sign language to text and voice



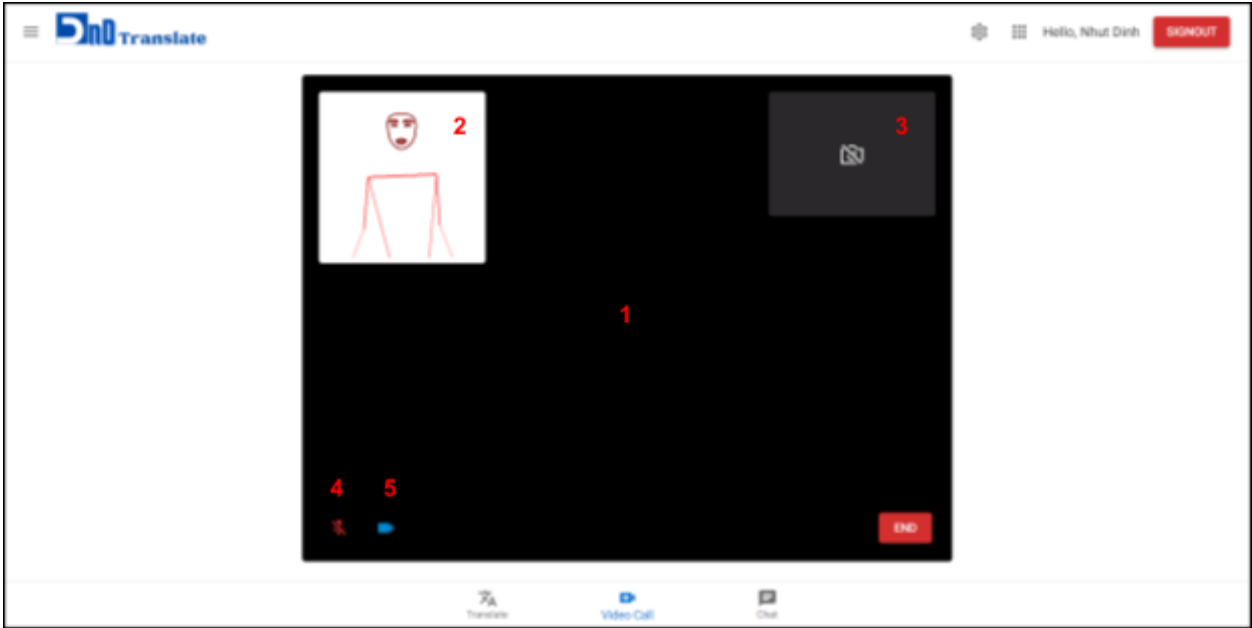
ID	Condition	Control Type	Target	Notes
1	Click	Button	Click to start recording video	
2	Click	Button	Click to stop recording video	
3	Nothing	Image	Show video user	
4	Nothing	TextField	Show text predicted from sign language	
5	Click	Button	Make sound to text	

3.3.4. Convert voice to sign language during video calls



ID	Condition	Control Type	Target	Notes
1	Nothing	Image	Show video of other user	
2	Nothing	SignLanguage	Show sign language of	
3	Nothing	Image	Show video of self	
4	Click	Button	On/Off speak	
5	Click	Button	On/Off webcam	

3.3.5. Convert sign language to voice during video calls



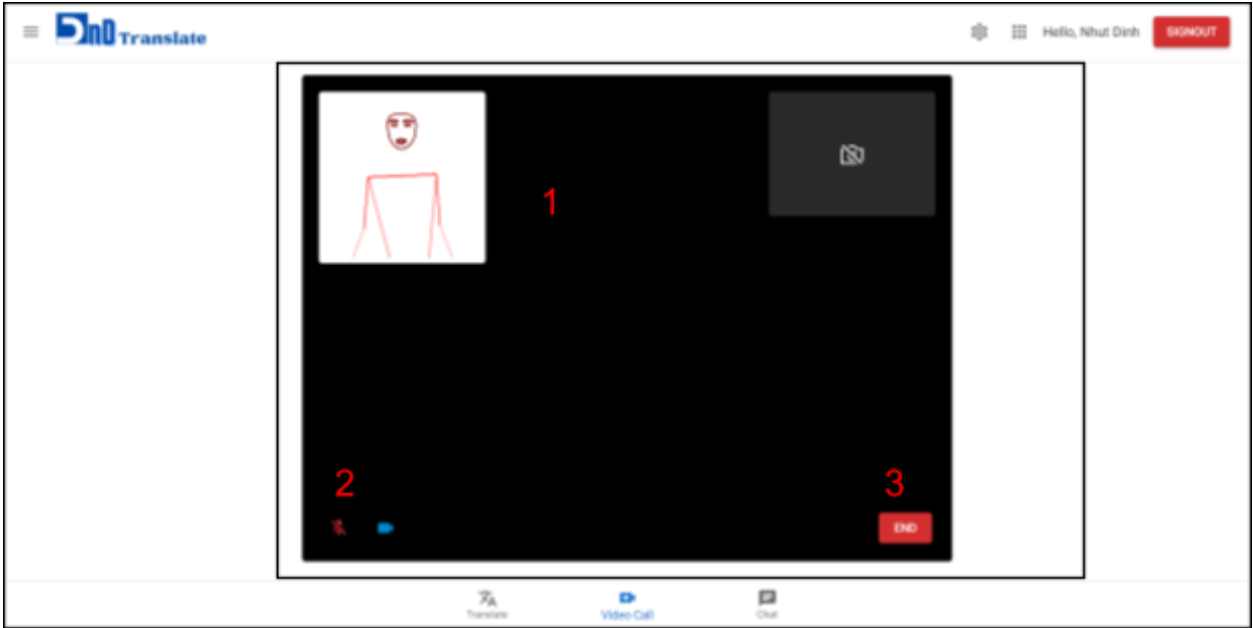
ID	Condition	Control Type	Target	Notes
1	Nothing	Image	Show video of other user	
2	Nothing	SignLanguage	Show sign language of	
3	Nothing	Image	Show video of self	
4	Click	Button	On/Off speak	
5	Click	Button	On/Off webcam	

3.4. Video Calls and Conferencing

3.4.1. Create a room for video calls

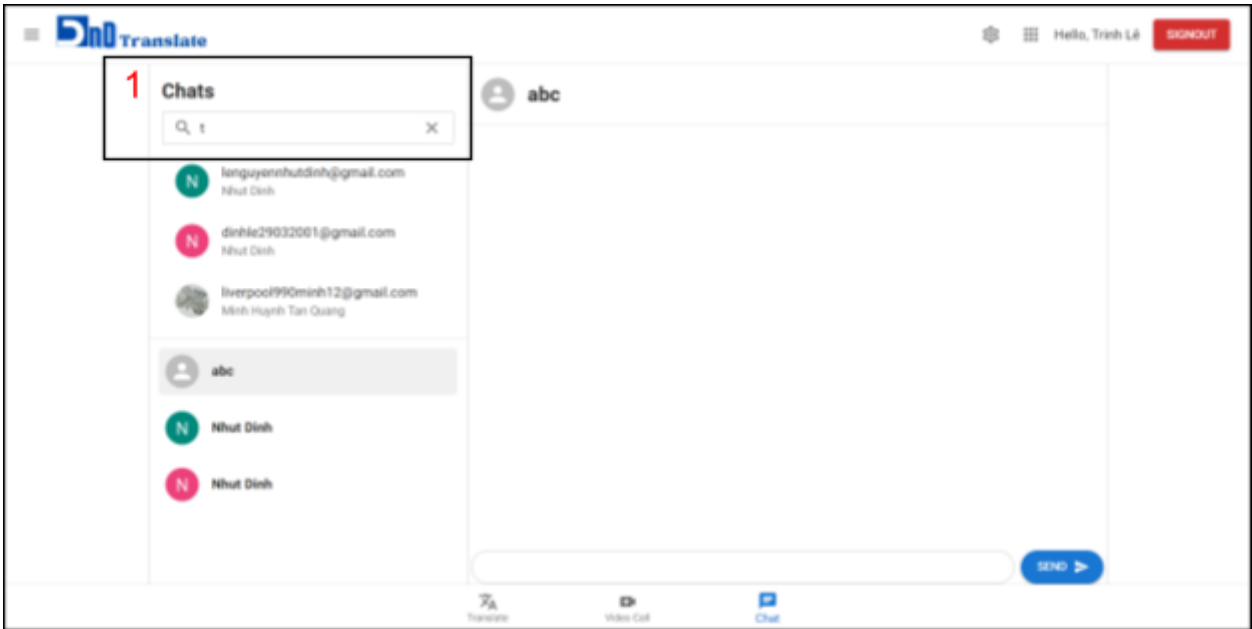
ID	Condition	Control Type	Target	Notes
1	Click	TextField	Fill channel name	
2	Click	RadioButton	Choose user type	
3	Click	Button	Click join to redirect to Video call room	

3.4.2. Leave a room



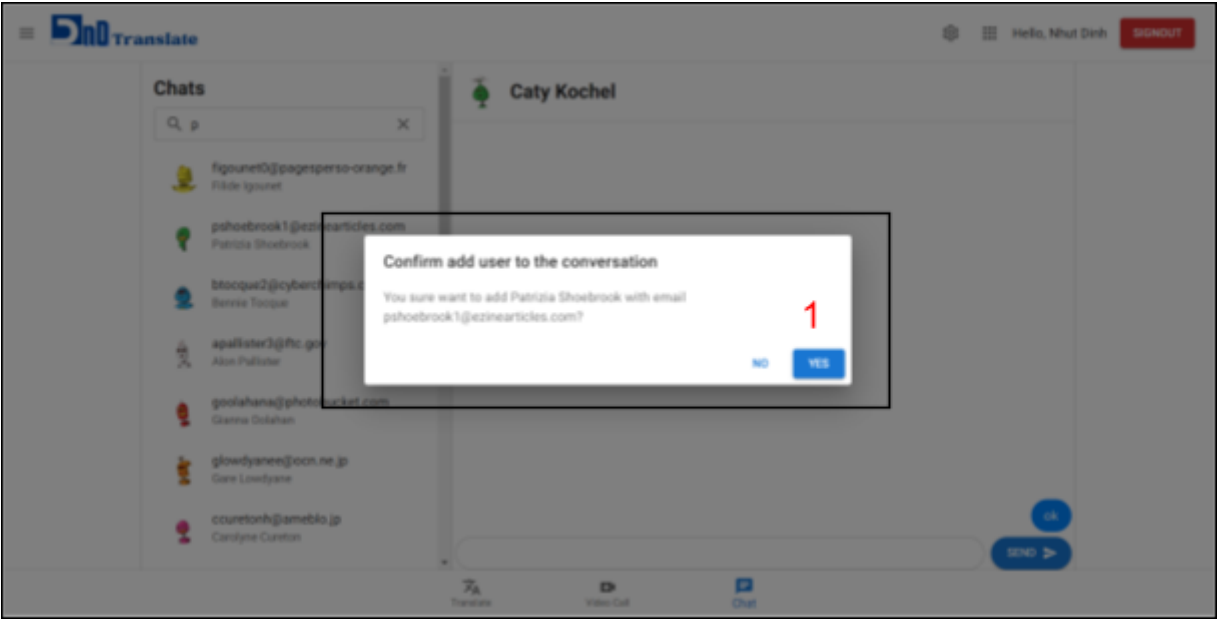
ID	Condition	Control Type	Target	Notes
1	Click	Button	Show action with sign language	
2	Click	Button	Speech or non-speech when call	
3	Click	Button	End call	

3.4.3. Search for other users within the platform



ID	Condition	Control Type	Target	Notes
1	Click	Button	Find friend in list friend	

3.4.4. Add other users to a conversation

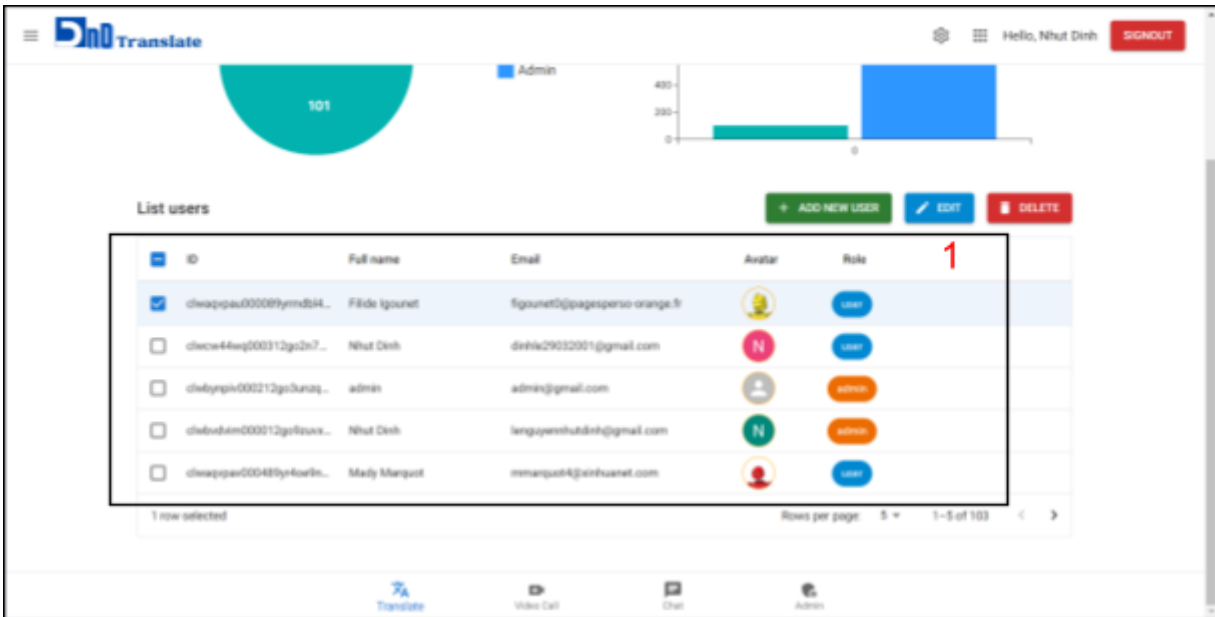


ID	Condition	Control Type	Target	Notes
1	Click	Button	Add or no add friend	

3.5. User Management and Administration

3.5.1. Send email invitations to other user

3.5.2. List all users registered on the platform



ID	Condition	Control Type	Target	Notes
1	Click	Button	Show list all Users	